

Worksheet 1: Decision Trees

Hamda Hassan

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1. Load packages

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(rpart)
library(rpart.plot)
```

```
## Warning: package 'rpart.plot' was built under R version 4.5.2
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(randomForest)
```

```
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
```

```
##      combine
##
## The following object is masked from 'package:ggplot2':
##
##      margin
```

```
library(gbm)
```

```
## Warning: package 'gbm' was built under R version 4.5.2
```

```
## Loaded gbm 2.2.3
```

```
## This version of gbm is no longer under development. Consider transitioning to gbm3, https://github.com
```

```
set.seed(123)
```

2. Load data

```
load("youth_data.Rdata")
data <- df
```

3. Outcomes

```
# Binary outcome (FOR TREE + RF)
data$mj_binary_factor <- ifelse(data$MRJFLAG == 1, "Yes", "No")
data$mj_binary_factor <- as.factor(data$mj_binary_factor)

# Binary outcome (FOR GBM)
data$mj_binary_num <- ifelse(data$MRJFLAG == 1, 1, 0)
data$mj_binary_num <- as.numeric(data$mj_binary_num)

# Regression outcome
data$mj_days <- data$MRJYDAYS
```

4. Predictors

```
predictors <- c(
  "IRSEX", "NEWRACE2", "INCOME", "POVERTY3",
  "IMOTHER", "IFATHER", "PARCHKHW", "PARHLPWH",
  "FRDMEVR2", "FRDMJMON", "PRMJJEVR2",
  "TALKPROB", "PREVIOL2", "ANYEDUC3"
)
```

5. Model dataset

```
# Dataset for TREE + RANDOM FOREST
model_data_class <- data[, c(
  predictors,
  "mj_binary_factor"
)]
```

```
# Dataset for GBM
model_data_gbm <- data[, c(
  predictors,
  "mj_binary_num"
)]
```

data cleaning

```
model_data_class <- na.omit(model_data_class)
model_data_gbm <- na.omit(model_data_gbm)
```

6. Train/Test split

```
# Classification split
index_class <- createDataPartition(model_data_class$mj_binary_factor,
                                   p = 0.7,
                                   list = FALSE)

train_class <- model_data_class[index_class, ]
test_class <- model_data_class[-index_class, ]

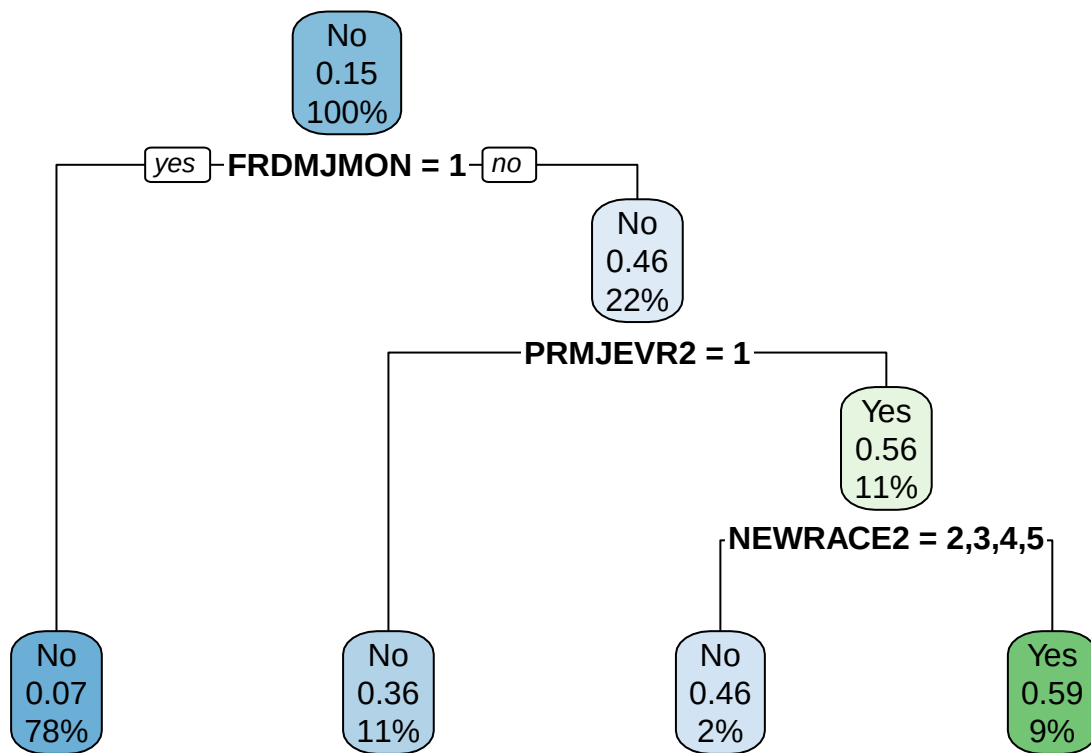
# GBM split
index_gbm <- createDataPartition(model_data_gbm$mj_binary_num,
                                 p = 0.7,
                                 list = FALSE)

train_gbm <- model_data_gbm[index_gbm, ]
test_gbm <- model_data_gbm[-index_gbm, ]
```

7. Decision tree

```
tree1 <- rpart(mj_binary_factor ~ .,
              data = train_class,
              method = "class")

rpart.plot(tree1)
```



8. Random Forest

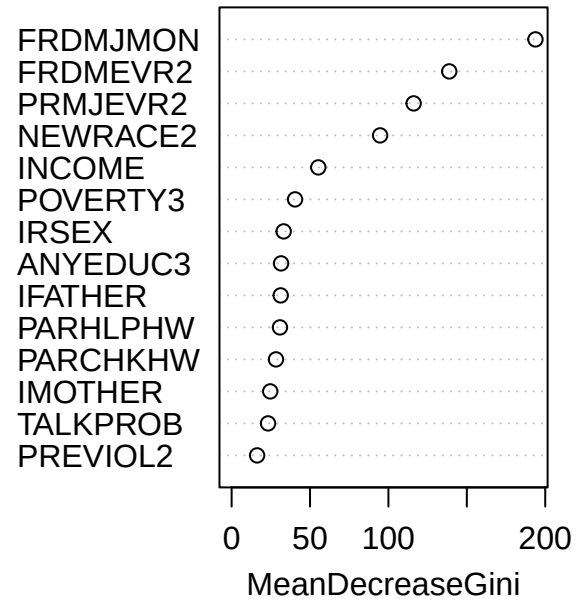
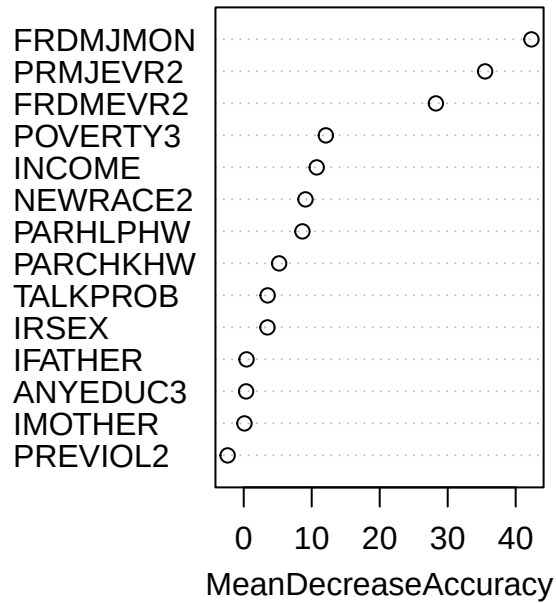
```

rf1 <- randomForest(mj_binary_factor ~ .,
  data = train_class,
  ntree = 300,
  importance = TRUE)

varImpPlot(rf1)

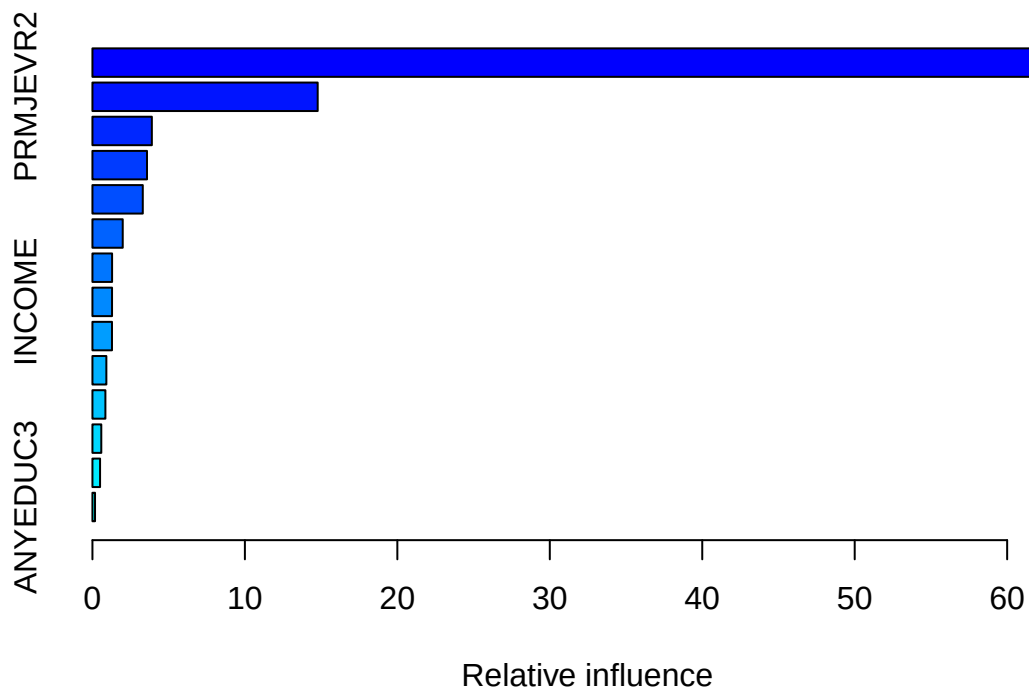
```

rf1



9. Boosting (GMB)

```
boost1 <- gbm(mj_binary_num ~ .,  
              data = train_gbm,  
              distribution = "bernoulli",  
              n.trees = 500,  
              interaction.depth = 3,  
              shrinkage = 0.01)  
  
summary(boost1)
```



```
##          var      rel.inf
## FRDMJMON FRDMJMON 65.5894602
## PRMJJEVR2 PRMJJEVR2 14.7751672
## NEWRACE2 NEWRACE2  3.8976842
## FRDMEVR2 FRDMEVR2  3.5793785
## PARHLPHW PARHLPHW  3.3033023
## IRSEX      IRSEX   1.9866163
## IMOTHER    IMOTHER 1.2886513
## INCOME      INCOME 1.2852880
## POVERTY3    POVERTY3 1.2816385
## PARCHKHW    PARCHKHW 0.9125230
## TALKPROB    TALKPROB 0.8502529
## PREVIOL2    PREVIOL2 0.5799924
## IFATHER     IFATHER 0.5005460
## ANYEDUC3    ANYEDUC3 0.1694992
```

10. Evaluation (Random Forest)

```
pred <- predict(rf1, test_class, type = "class")
confusionMatrix(pred, test_class$mj_binary_factor)
```

```
## Confusion Matrix and Statistics
##
##          Reference
```

```
## Prediction    No  Yes
##           No 2401 309
##           Yes 107 137
##
##           Accuracy : 0.8592
##           95% CI : (0.8461, 0.8715)
##           No Information Rate : 0.849
##           P-Value [Acc > NIR] : 0.06375
##
##           Kappa : 0.325
##
## Mcnemar's Test P-Value : < 2e-16
##
##           Sensitivity : 0.9573
##           Specificity : 0.3072
##           Pos Pred Value : 0.8860
##           Neg Pred Value : 0.5615
##           Prevalence : 0.8490
##           Detection Rate : 0.8128
##           Detection Prevalence : 0.9174
##           Balanced Accuracy : 0.6323
##
##           'Positive' Class : No
##
```

Check for NAs

```
colSums(is.na(model_data_class))
```

```
##           IRSEX           NEWRACE2           INCOME           POVERTY3
##           0           0           0           0
##           IMOTHER           IFATHER           PARCHKHW           PARHLPHW
##           0           0           0           0
##           FRDMEVR2           FRDMJMON           PRMJEV2           TALKPROB
##           0           0           0           0
##           PREVIOL2           ANYEDUC3 mj_binary_factor
##           0           0           0
```

```
colSums(is.na(model_data_gbm))
```

```
##           IRSEX           NEWRACE2           INCOME           POVERTY3           IMOTHER
##           0           0           0           0           0
##           IFATHER           PARCHKHW           PARHLPHW           FRDMEVR2           FRDMJMON
##           0           0           0           0           0
##           PRMJEV2           TALKPROB           PREVIOL2           ANYEDUC3 mj_binary_num
##           0           0           0           0           0
```